PCRQ 9

Graded

Student

VK Nguyen

Total Points

3 / 4 pts

Question 1

(no title) **1** / 1 pt

✔ **+ 1 pt** Correct

+ 0 pts Incorrect

Question 2

(no title) **0** / 1 pt

+ 1 pt Correct

✔ **+ 0 pts** Incorrect

Question 3

(no title) **1** / 1 pt

✔ **+ 1 pt** Correct

+ 0 pts Incorrect

Question 4

(no title) **1** / 1 pt

✔ **+ 1 pt** good faith answer

+ 0 pts blank

The interval $(u^* - \varepsilon, u^*]$ is the set of all numbers x with
\[
u^* - \varepsilon < x \le u^*.
\]
The ``$('' means $u^* - \varepsilon$ is left out, and the ``$]$'' means u^* is included.
In the picture, it is the stretch under the curly brace: a small window just below u^*.

The statement $S \cap (u^* - \varepsilon, u^*] \neq \varnothing$ says that some element of S
lies inside that window. That is the dot labeled $x \in S$.

Why is it there? Nothing in S is past u^*, since u^* is an upper bound.
But $u^* - \varepsilon$ is \emph{not} an upper bound, so some element of S is past
$u^* - \varepsilon$. That element is trapped in the window.

So the picture says: however thin you make the window below the supremum, some point of S always lands in it.

Q1
1 Point

Let $S \subseteq \mathbb{R}$ be nonempty and upper bounded, and suppose that u is an upper bound of S. Which additional condition guarantees that $u = \sup(S)$?

- ○ There exists $x \in S$ such that $x > u - 1$.
- ○ There exists $\varepsilon > 0$ such that every $x \in S$ satisfies $x \geq u - \varepsilon$.
- ◉ For every $\varepsilon > 0$, there exists $x \in S$ such that $x > u - \varepsilon$.
- ○ Every element of S is strictly less than u.

Q2
1 Point

Which statement is true?

- ◉ $\exists n \in \mathbb{N}\ \forall x \in \mathbb{R}\ n > x$.
- ○ $\forall x \in \mathbb{R}\ \exists n \in \mathbb{N}\ n > x$.

Q3
1 Point

Consider the set $S = \left\{ \frac{n}{n+1} : n \in \mathbb{N} \right\}$.
Which statement is correct?

- ○ The set has maximum 1.
- ◉ The set has supremum 1 but it has no maximum.
- ○ The set is not upper bounded.
- ○ The set has supremum $\frac{1}{2}$.

Q4

1 Point

Please write briefly (in two or three sentences) what single point of the notes of Lecture 10 you found to be the most difficult or confusing. If you did not find any part of it difficult or confusing, please say what part(s) you found most interesting.

I'm confused about the geometric meaning of proposition 2.1. There is a half closed bracket in a description but an interval illustrated on a line is just confusing me. In simple terms, I just don't get how the illustration illustrate the description just above it.